Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-Enabled Intelligence Wearable Robotic Assistance for Human Mobility			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (), Advanced Mobility (√), Next-gen Nuclear Power (), Advanced Bio (√), Aerospace/Marine (), Hydrogen (), Cybersecurity (), AI (√), Next-gen Communication (), Advanced Robotics & Manufacturing (√), Quantum ()			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical · Materials		Bio · Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/ Automation Machinery	Robot Control and Intelligence Technologies	*Functional Rehabilitation/Assistance and Welfare Devices*	Mobility Assistance Devices
Concept of Proposed Technology	The proposed technology aims to develop an **AI-driven wearable robotic assistive system** that enhances human mobility. Unlike conventional wearable robots that rely on task-specific control strategies, this system leverages advanced artificial intelligence to infer human kinetic and kinematic states directly from wearable sensing modalities.By estimating biomechanical variables in real time, the system enables task-agnostic assistance that can adapt across diverse locomotor activities, including walking, incline, stair, and other community ambulation tasks. This approach facilitates the translation of wearable robotic technology from controlled laboratory settings to			

(max 10 pages)

	practical daily-life use. Moreover, the proposed framework is extensible across user populations, supporting human augmentation for able-bodied users as well as assistive and rehabilitative applications for clinical populations and older adults, thereby offering strong commercial scalability across healthcare, wellness, and consumer wearable markets. Key features: ● AI-enabled real-time estimation of human kinetic and kinematic states using wearable sensing ● Task-agnostic assistance framework adaptable to multiple locomotor tasks and environments ● Platform-independent control architecture applicable across different wearable robotic systems
Rationale for Joint R&D Project	Successful deployment of AI-driven wearable robotics in real-world environments requires close collaboration between research institutions and industrial partners. While research institutes provide expertise in AI-based modeling, biomechanics, and control algorithms, industrial partners contribute essential capabilities in hardware design, system integration, and manufacturability. A joint R&D framework is critical to co-design AI algorithms and wearable robotic hardware under realistic constraints, such as size, weight, power consumption, safety, and robustness. Rationale for collaboration: ● The research institute contributes expertise in AI-based biomechanical state estimation and control algorithms ● The Korean SME partner contributes wearable robot hardware development, system integration, and validation capabilities ● Joint optimization of software and hardware is essential to achieve reliable and deployable wearable assistive devices Expected benefits of joint R&D: ● Accelerated transition from AI research to wearable robotic prototypes ● Reduced technical risk through iterative hardware–software co-development ● Establishment of a scalable foundation for commercialization of wearable assistive robotics
Key Words	1. AI-Driven Wearable Robotics 2. Task-Agnostic Mobility Assistance 3. Real-Time Biomechanical State Estimation 4. Human–Robot Interaction 5. Wearable Assistive Devices

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	The objective of this project is to develop an AI-driven wearable robotic system capable of providing reliable, task-agnostic mobility assistance. The specific R&D objectives are: ● Development of AI-based algorithms for real-time estimation of human kinetic and kinematic states from wearable sensor data ● Design of task-agnostic control strategies that generalize across diverse locomotor tasks ● Experimental validation of system performance under human–robot interaction conditions			
R&D Target Specifications	The target performance specifications of this R&D project are as follows: **Offline Biomechanical Estimation Performance (80%):** Achieve high-accuracy offline estimation of human biomechanical states (e.g., biological joint moments) using wearable sensor data, demonstrating robust performance across multiple locomotor tasks *($R^2 > 0.80$)*. **Online Accuracy and Assist Validation (80%):** Demonstrate accurate real-time estimation of biomechanical states during human–robot interaction, achieving performance comparable to offline estimation *($R^2 > 0.80$)* while enabling stable and transparent assistive control. **Experimental Validation of Biomechanical Benefits:** Experimentally demonstrate measurable biomechanical benefits resulting from the proposed assistance, including reductions in joint-level mechanical power and/or metabolic cost during locomotion.			

R&D Plan	This R&D project will be conducted over three years, following a progressive development and validation strategy from laboratory environments to real-world domestic settings.
	
	● Year 1: Algorithm Development and Validation
	  ○ Develop AI models for biomechanical state estimation using wearable sensor data
	  ○ Validate estimation and control algorithms in benchtop environments
	● Year 2: Laboratory-Scale Hardware Integration and Human Testing
	  ○ Integrate AI algorithms with wearable robotic hardware
	  ○ Conduct controlled human experiments to evaluate accuracy, stability, and robustness
	  ○ Optimize algorithms for real-time embedded implementation
	● Year 3: Real-World Deployment and Performance Validation
	  ○ Deploy the wearable robotic system in community ambulation scenarios
	  ○ Evaluate system performance across diverse real-world tasks
	  ○ Refine system reliability, safety, and usability for technology transfer			

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will be responsible for the development of core AI algorithms and system-level intelligence for the proposed wearable robotic assistance technology. The institute will lead the research on real-time biomechanical state estimation, human–robot interaction, and control methodologies required to achieve reliable and adaptive mobility assistance. The specific roles and responsibilities of the research institute include: ● Development of AI-based biomechanical estimation and control algorithms ● Design of system architecture and data-driven control framework ● Simulation, analysis, and experimental evaluation of algorithm performance ● Technical leadership and coordination with the industrial partner Technical guidance and collaboration with the industrial partner to support system integration and experimental evaluation
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for the development, integration, and validation of the wearable robotic hardware and system-level implementation required for the proposed technology. The SME will play a key role in translating AI-based estimation and control algorithms into a practical, safe, and deployable wearable robotic system by addressing hardware constraints, embedded implementation, and real-world usability considerations. The specific roles and responsibilities of the Korean SME partner include: ● Design and development of wearable robotic hardware ● System integration of AI algorithms with sensors and embedded controllers ● Execution of laboratory and real-world experiments ● Contribution to reliability improvement and commercialization planning

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	The proposed project is expected to deliver a validated AI-driven wearable robotic prototype capable of task-agnostic mobility assistance in real-world environments. The project will advance the state of the art in AI-based human state estimation and wearable robot control, enabling intuitive and robust human–robot interaction. From an industrial perspective, the outcomes will provide the Korean SME partner with core AI-enabled wearable robotics technologies applicable to next-generation assistive devices. More broadly, this project will contribute to the growth of the wearable robotics industry and strengthen international collaboration in AI-driven assistive technologies.

6. References

(1) Molinaro, D. D., Scherpereel, K. L., Schonhaut, E. B., Evangelopoulos, G., Shepherd, M. K., & Young, A. J. (2024). Task-agnostic exoskeleton control via biological joint moment estimation. *Nature*, *635*(8038), 337-344.

(2) Molinaro, D. D., Kang, I., & Young, A. J. (2024). Estimating human joint moments unifies exoskeleton control, reducing user effort. *Science robotics*, *9*(88), eadi8852.

(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

(max 10 pages)